higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 4

(12041034)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

Copyright reserved

QUESTION 1

1.1 The South African power grid is mostly supplied with electricity generated by coal-fired power stations.

Name THREE other forms of power-generating sources that supply the national power grid with electricity. (3)

1.2 State the function of each of the following components used in the generation, transmission and distribution of electricity:

1.2.1 Switchgear

1.2.2 Electrostatic precipitator

(2 × 2) (4)

1.3 An alternator must first be synchronised with the power grid before it can be connected to it.

Which THREE conditions should be met before an alternator can be connected to a power grid? (3 × 2) (6)

1.4 Give TWO advantages of a three-phase four-wire system in a small-town power grid. (2 × 2) (4)

1.5 Give THREE faults that normally occur on power-transmission lines. (3)

[20]

QUESTION 2

2.1 Explain the term *solenoid.* (2)

2.2 State the function of each of the following components:

2.2.1 Overload relays

2.2.2 Limit switches

(2 × 2) (4)

2.3 Draw a star-delta main circuit for a three-phase 380/400 V motor. (7)

2.4 One of the test instruments used for finding a fault on an underground cable is the thumper tester.

Explain the procedure when using a thumper or pulse generator to find a fault on a cable. (2)

[15]

Copyright reserved

QUESTION 3

3.1 List THREE pieces of information to gather before marking out the route for the construction of overhead power lines. (3)

3.2 Explain how to connect a stay wire to a pole. (2)

3.3 Define *risk assessment* and explain how it helps to ensure safety during the planning stages (2)

3.4 List THREE factors to consider when spacing poles for transmission lines. (3)

3.5 After a three-phase medium-voltage overhead supply is constructed, testing and commissioning are done.

Give FOUR commissioning tests done on a completed overhead supply line. (4)

3.6 Name the type of conductor most commonly used for the construction of a medium-voltage overhead supply. (1)

[15]

QUESTION 4

4.1 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–N) next to the question number (4.1.1–4.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.1.1	Ensures measuring instruments give accurate readings	A	electric shock
4.1.2	Electrical continuity and mechanically sound	B	wiring diagram
4.1.3	Part of a switchyard	C	hipot tester
4.1.4	Removal of jewellery when working on live circuits	D	hazard identification
4.1.5	Testing possible on such an installation	E	highest voltage
4.1.6	Risk assessment at the workplace	F	5%
4.1.7	Conduct insulation resistance on 11 kV cable	G	10%
4.1.8	Show the location and type of protection devices to be used	H	terminations
4.1.9	Rating on insulating gloves	I	one-line schematic diagram
4.1.10	Voltage drop from the main point of supply to the furthest point of consumption	J	live
		K	dead
		L	calibration
		M	megger
		N	arcing horn

(10 × 1) (10)

4.2 After an installation has been wired, tests must be conducted.

Explain how to conduct a voltage-on-load test. (4)

4.3 Give ONE reason for discharging stored energy on components. (2)

4.4 Name TWO legislations to consult when working with low-voltage electrical installations. (2)

4.5 A certificate of compliance (COC) has a section for comments.

Which details should appear in the section for comments? (2)

[20]

Copyright reserved

QUESTION 5

5.1 An abnormal condition has been detected within the tank of a transformer.

Explain how a transformer can be protected from this abnormal condition. (2)

5.2 A rotor has been removed from a stator of a three-phase electric motor. As an electrician you are called to do maintenance on the motor.

Give TWO components of the motor that you will do maintenance on. (2)

5.3 An electric motor is running. After a while an oil-dashpot overload relay trips (assume there is no fault on the motor).

Give TWO causes and TWO corrective actions for a tripping oil-dashpot overload relay. (2 + 2) (4)

5.4 An underground cable is suspected to be faulty.

List THREE types of techniques/methods to locate an underground cable fault. (3)

5.5 An AC-voltage tester must be used to determine which side of a transformer is faulty or open circuited.

Explain how to perform this test.(Assume there is no short circuit in winings) (4)

[15]

QUESTION 6

6.1 Briefly explain the *indirectly connected wind-turbine system*. (3)

6.2 Mr Zwane owns a two-roomed house and wants to shift the power consumed by the following appliances to a solar system:

APPLIANCE AND VOLTAGE SUPPLY	POWER CONSUMED PER DAY
Fridge (230 V)	5,04 kWh
TWO Lights (230 V)	0,012 kWh

Calculate how much he will save on electricity per month when the cost per kilowatt-hour is R1,25. Assume 30 days per month. (4)

Copyright reserved

6.3 The following details apply to a battery:

Capacity:	2 Ah
Voltage:	24 V
Daily power usage:	1,6 kWh
Battery discharge depth:	70% (ignore temperature capacity or useful adjustment)

Determine for how many days the battery bank will supply the load. (4)

6.4 Give TWO activities that must be done or identified during the survey that must be conducted before installing a solar panel. (2)

6.5 Explain the function of a solar regulator in a solar-power system. (2)
[15]

TOTAL: 100

Copyright reserved